



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam: : Trimester: Fall 2025

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 2 hours Total Marks: 40

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Write your serial number of exam attendance sheet on your answer script.

- 1 a) Show **manual tracing** for the following program and find **output**. 5

```
#include<stdio.h>
int a;
float func1(int x);
void func2(float x, int y);
int main(){
    a=11;
    printf("%f\n", func1(a+50));
    func2(6.5, 10);
    printf("%d\n", a);
    return 0;
}
float func1(int x) {
    a=x*10;
    printf("%d\n", x+25);
    return x+a+20.5;
}
void func2(float x, int y){
    printf("%f\n", x+5);
    a=a+y;
}
```

- b) Write a **program** using **two user defined functions** to perform the following operations 5

- i) The function **main()** reads two integer numbers from keyboard and pass the numbers to the **CalValue()** function.
- ii) The user defined function **Differ(int x, int y)** checks whether the difference of x and y is positive or negative. If it is positive, it will return **1**, otherwise **0**.
- iii) The user defined function **CalValue(int x, int y)** calculates the value of $(x+y)(x-y)$ if the **Differ()** function returns 1, and otherwise calculates the value of $(y+x)(y-x)$. This function returns the calculated value to the calling function.
- iv) The **main()** function prints the returned value from the function **CalValue()**.

- 2 a) Show **manual tracing** for the following program 4

```
char Str1[13]={'\0'};
char Str2[4]={'\0'};
int i, j, k;
strcpy(Str1, "CSE DEPT ");
strcpy(Str2, "UIU");
i=strlen(Str1);
printf("%s %s %d\n", Str1, Str2, i);
k=0;
for(j=i; j<i+strlen(Str2); j++){
    Str1[[j]=Str2[k++];
}
printf("%s %s %d\n", Str1, Str2, j);
```

```

if (strcmp(Str2, Str1)<0)
    printf("String %s is greater", Str1);
else
    printf("String %s is greater", Str2);

```

- b) Show **manual tracing** for the following program and find **output**.

4

```

#include<stdio.h>

void func(int a, int *b){
    int *q;
    q=b;
    *q=*q+a;
    a=a-3;
}

int main(){
    int a=5, b=9;
    printf("%d %d\n", a+b, b);
    func(a, &b);
    printf("%d %d", a+b, b);
    return 0;
}

```

- 3 a) Show **manual tracing** for the following program and find **output**.

5

```

#include<stdio.h>
void func(int B[], int n);
int main(){
    int A[5]={10, 20, 30, 40, 50};
    int i;
    for(i=0; i<5; i++)
        printf("%d ", --A[i]);
    func(A, 5);
    printf("\n");
    for(i=0; i<5; i++)
        printf("%d ", ++A[i]);
    return 0;
}

void func(int B[], int n){
    int i;
    for(i=0; i<n; i++){
        B[i]=B[i]%3+1;
    }
}

```

- b) i) Distinguish between write and append modes in file writing.
 ii) What type of problems will be happened if a file is not closed after opening?
 iii) What are the uses and syntax of the following standard functions?
 fopen(), fprintf(), fscanf(), fclose()

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- 4 Write a **program** having the structure **student (name, id, course-I marks, course-II marks, phone, email)** to perform the following operations for two students

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- a) Read name, id, course-I marks, course-II marks, phone and email of two students from keyboard
- b) Find the average of the marks of the two courses for each student
- c) Display the following sample report on monitor for these two students:

Rahim 10 80.0 90.0 01632078812 xxx@gmail.com
Saiham 20 70.0 80.0 01710012300 yyy@hotmail.com

Average Marks of Rahim is 85.5
Average Marks of Saiham is 75.5

5 Write a **code segment** to reverse an input string without using any string related standard function(s). 4

Input String="CSE Dept"
Output String="tpeD ESC"